

Computer Networks Lab Report: Network Traffic and Protocol Analysis

Course: Computer Networks Lab (Assignment 6)

Date: October 17, 2025

Topic: Network Traffic Capture and Protocol Analysis using Wireshark

1. Objective

The objective of this assignment was to capture and analyze live network traffic using Wireshark. This involved identifying various network protocols, interpreting packet-level details such as headers and fields, and understanding core communication patterns through filtering and analysis.

2. Task 1: Packet Capture

A 2-minute packet capture was initiated on the active Wi-Fi interface. During the capture, traffic was generated by pinging 8.8.8.8 and visiting several websites (<http://example.com>, <https://wikipedia.org>). The resulting data was saved as `capture_lab1.pcap`.

3. Task 2: Protocol Identification

Display filters were used to isolate specific protocols from the main capture file.

3.1. ICMP Traffic

The icmp filter successfully isolated the "Echo (ping) request" and "Echo (ping) reply" packets generated by the ping 8.8.8.8 command.

The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. The main window is divided into three panes:

- Packet List Pane:** Displays a list of captured packets. A filter 'icmp' is applied. The list shows packets 4439 through 4588. Packet 4532 is selected, highlighted in red. The columns are No., Time, Source, Destination, Protocol, Length, and Info.
- Packet Details Pane:** Shows the hierarchical structure of the selected packet (4532). It includes Ethernet II, Internet Protocol Version 4, and Internet Control Message Protocol.
- Packet Bytes Pane:** Displays the raw data of the selected packet in hexadecimal and ASCII format.

The status bar at the bottom indicates 'Packets: 8033 · Displayed: 8 (0.1%)' and 'Profile: Default'. The system tray at the bottom right shows the date and time as 15:04 on 17-10-2025.

No.	Time	Source	Destination	Protocol	Length	Info
4439	78.174381	10.120.44.171	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=17/4352, ttl=128 (re
4442	78.267219	8.8.8.8	10.120.44.171	ICMP	78	Echo (ping) reply id=0x0001, seq=17/4352, ttl=113 (re
4495	79.182717	10.120.44.171	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=18/4608, ttl=128 (re
4500	79.299095	8.8.8.8	10.120.44.171	ICMP	78	Echo (ping) reply id=0x0001, seq=18/4608, ttl=113 (re
4532	80.205267	10.120.44.171	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=19/4864, ttl=128 (re
4540	80.308849	8.8.8.8	10.120.44.171	ICMP	78	Echo (ping) reply id=0x0001, seq=19/4864, ttl=113 (re
4581	81.214385	10.120.44.171	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=20/5120, ttl=128 (re
4588	81.299221	8.8.8.8	10.120.44.171	ICMP	78	Echo (ping) reply id=0x0001, seq=20/5120, ttl=113 (re

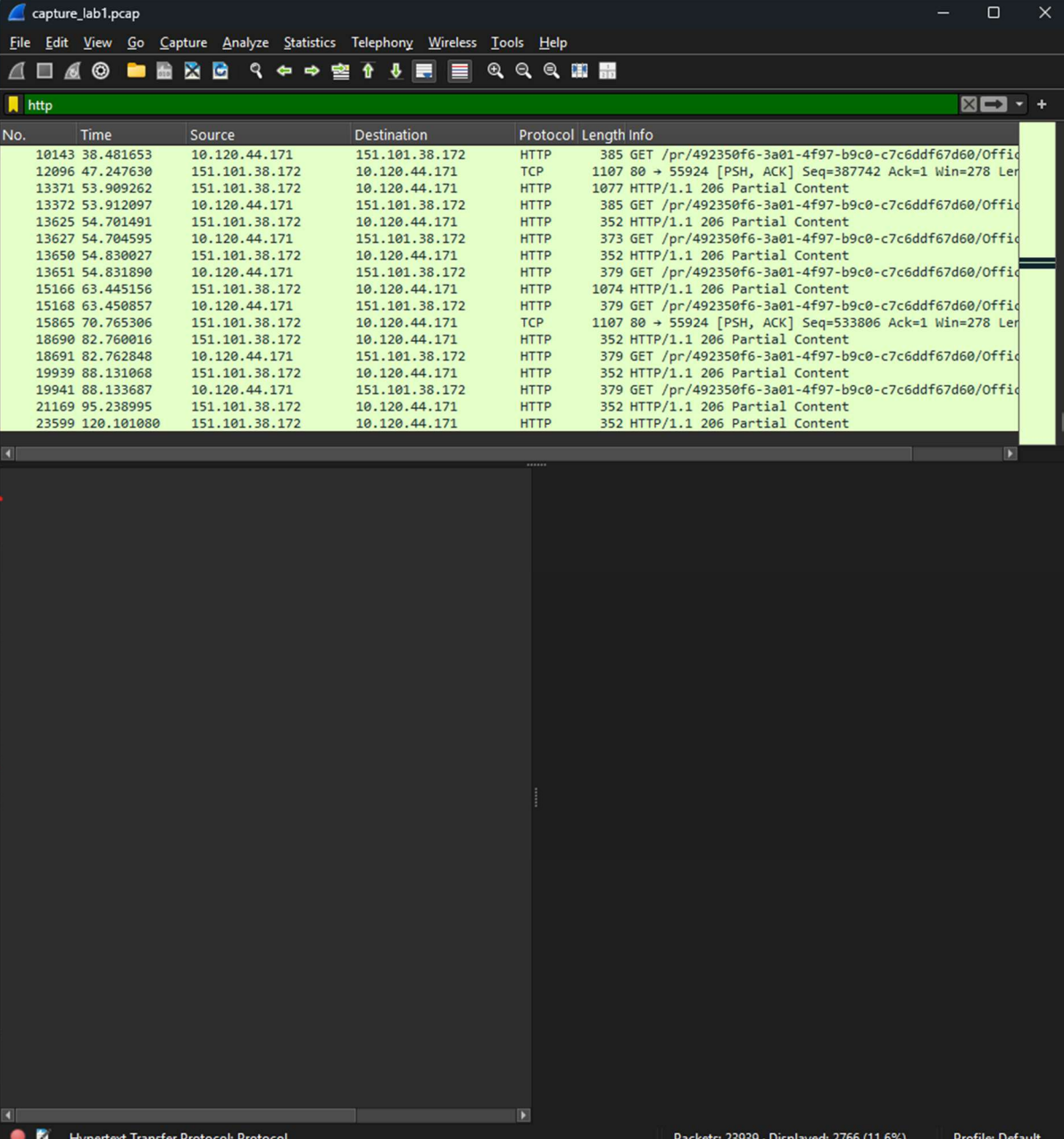
Frame 4532: Packet, 74 bytes on wire (592 bits), 74 bytes captured
Ethernet II, Src: Intel_8c:c1:b1 (d0:65:78:8c:c1:b1), Dst: 8a:d1:
Internet Protocol Version 4, Src: 10.120.44.171, Dst: 8.8.8.8
Internet Control Message Protocol

0000 8a d1 fe 27 22 5b d0 65 78 8c c1 b1 08 00 45 00 ..."[e x
0010 00 3c ae 75 00 00 80 01 00 00 0a 78 2c ab 08 08 <-u...
0020 08 08 08 00 4d 48 00 01 00 13 61 62 63 64 65 66 ...MH..
0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn o
0040 77 61 62 63 64 65 66 67 68 69 wabcdefg h

Internet Control Message Protocol: Protocol Packets: 8033 · Displayed: 8 (0.1%) Profile: Default

3.2. HTTP Traffic

The http filter isolated the unencrypted web traffic from visiting <http://example.com>. The GET request packet is clearly visible.



The screenshot shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The packet list pane displays a list of captured packets, with the following columns: No., Time, Source, Destination, Protocol, and Length Info. The packets are filtered by 'http' in the filter bar. The packet details pane shows the selected packet (No. 10143) and its details, including the Hypertext Transfer Protocol section. The packet bytes pane is currently empty.

No.	Time	Source	Destination	Protocol	Length Info
10143	38.481653	10.120.44.171	151.101.38.172	HTTP	385 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
12096	47.247630	151.101.38.172	10.120.44.171	TCP	1107 80 → 55924 [PSH, ACK] Seq=387742 Ack=1 Win=278 Len
13371	53.909262	151.101.38.172	10.120.44.171	HTTP	1077 HTTP/1.1 206 Partial Content
13372	53.912097	10.120.44.171	151.101.38.172	HTTP	385 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
13625	54.701491	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content
13627	54.704595	10.120.44.171	151.101.38.172	HTTP	373 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
13650	54.830027	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content
13651	54.831890	10.120.44.171	151.101.38.172	HTTP	379 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
15166	63.445156	151.101.38.172	10.120.44.171	HTTP	1074 HTTP/1.1 206 Partial Content
15168	63.450857	10.120.44.171	151.101.38.172	HTTP	379 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
15865	70.765306	151.101.38.172	10.120.44.171	TCP	1107 80 → 55924 [PSH, ACK] Seq=533806 Ack=1 Win=278 Len
18690	82.760016	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content
18691	82.762848	10.120.44.171	151.101.38.172	HTTP	379 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
19939	88.131068	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content
19941	88.133687	10.120.44.171	151.101.38.172	HTTP	379 GET /pr/492350f6-3a01-4f97-b9c0-c7c6ddf67d60/Offic
21169	95.238995	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content
23599	120.101080	151.101.38.172	10.120.44.171	HTTP	352 HTTP/1.1 206 Partial Content

Hypertext Transfer Protocol: Protocol

Packets: 23939 · Displayed: 2766 (11.6%) Profile: Default

3.3. DNS Traffic

The dns filter isolated the "Standard query" and "Standard query response" packets, which are responsible for resolving domain names (like example.com) into IP addresses.

capture_lab1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
22978	113.869816	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x66aa AAAA wikipedia.org
22980	113.869917	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xa7dc HTTPS wikipedia.org
22982	113.870240	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xc7ec A wikipedia.org
22984	113.870374	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xb8db HTTPS wikipedia.org
22986	113.870460	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x07a6 A wikipedia.org
22988	113.870541	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xc45 AAAA wikipedia.org
23035	114.293316	10.120.44.9	10.120.44.171	DNS	147	Standard query response 0xb8db HTTPS wikipedia.org
23036	114.293316	10.120.44.9	10.120.44.171	DNS	147	Standard query response 0xa7dc HTTPS wikipedia.org
23054	114.370006	10.120.44.9	10.120.44.171	DNS	114	Standard query response 0x66aa AAAA wikipedia.org
23055	114.370006	10.120.44.9	10.120.44.171	DNS	102	Standard query response 0xc7ec A wikipedia.org A 1
23056	114.370006	10.120.44.9	10.120.44.171	DNS	102	Standard query response 0x07a6 A wikipedia.org A 1
23057	114.370006	10.120.44.9	10.120.44.171	DNS	114	Standard query response 0xc45 AAAA wikipedia.org
23075	114.378552	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xca9c A wikipedia.org
23077	114.378606	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x82b9 HTTPS wikipedia.org
23079	114.378643	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x39f3 AAAA wikipedia.org
23117	114.729926	10.120.44.9	10.120.44.171	DNS	114	Standard query response 0x39f3 AAAA wikipedia.org
23118	114.729926	10.120.44.9	10.120.44.171	DNS	147	Standard query response 0x82b9 HTTPS wikipedia.org
23145	114.855863	10.120.44.9	10.120.44.171	DNS	102	Standard query response 0xca9c A wikipedia.org A 1
23168	114.905946	10.120.44.171	10.120.44.9	DNS	89	Standard query 0xa1f9 HTTPS www.wikipedia.org
23170	114.906123	10.120.44.171	10.120.44.9	DNS	89	Standard query 0x7892 A www.wikipedia.org
23172	114.906198	10.120.44.171	10.120.44.9	DNS	89	Standard query 0xa318 AAAA www.wikipedia.org
23221	115.341292	10.120.44.9	10.120.44.171	DNS	147	Standard query response 0xa318 AAAA www.wikipedia.org
23228	115.404427	10.120.44.9	10.120.44.171	DNS	135	Standard query response 0x7892 A www.wikipedia.org
23241	115.487735	10.120.44.9	10.120.44.171	DNS	170	Standard query response 0xa1f9 HTTPS www.wikipedia.org
23273	115.736492	10.120.44.171	10.120.44.9	DNS	92	Standard query 0xc11d HTTPS upload.wikimedia.org
23279	115.737538	10.120.44.171	10.120.44.9	DNS	92	Standard query 0x8a45 AAAA upload.wikimedia.org
23281	115.737582	10.120.44.171	10.120.44.9	DNS	92	Standard query 0x6f65 A upload.wikimedia.org
23306	115.952783	10.120.44.9	10.120.44.171	DNS	121	Standard query response 0x8a45 AAAA upload.wikime
23312	115.968119	10.120.44.9	10.120.44.171	DNS	144	Standard query response 0xc11d HTTPS upload.wikime
23313	115.968119	10.120.44.9	10.120.44.171	DNS	109	Standard query response 0x6f65 A upload.wikimedia.
23338	116.119764	10.120.44.171	10.120.44.9	DNS	88	Standard query 0x6304 A en.wikipedia.org
23340	116.119821	10.120.44.171	10.120.44.9	DNS	88	Standard query 0x7d45 HTTPS en.wikipedia.org
23342	116.119859	10.120.44.171	10.120.44.9	DNS	88	Standard query 0x85ce AAAA en.wikipedia.org
23364	116.327020	10.120.44.9	10.120.44.171	DNS	134	Standard query response 0x6304 A en.wikipedia.org
23365	116.327020	10.120.44.9	10.120.44.171	DNS	146	Standard query response 0x85ce AAAA en.wikipedia.c
23366	116.327020	10.120.44.9	10.120.44.171	DNS	169	Standard query response 0x7d45 HTTPS en.wikipedia.c
23474	118.695608	10.120.44.171	10.120.44.9	DNS	99	Standard query 0x550c HTTPS safebrowsing.googleapis
23476	118.695759	10.120.44.171	10.120.44.9	DNS	99	Standard query 0xbb10 AAAA safebrowsing.googleapis
23478	118.695873	10.120.44.171	10.120.44.9	DNS	99	Standard query 0xa7a3 A safebrowsing.googleapis.co
23499	118.910046	10.120.44.9	10.120.44.171	DNS	128	Standard query response 0xbb10 AAAA safebrowsing.g
23505	118.926469	10.120.44.9	10.120.44.171	DNS	157	Standard query response 0x550c HTTPS safebrowsing.
23506	118.926469	10.120.44.9	10.120.44.171	DNS	116	Standard query response 0xa7a3 A safebrowsing.goog
23680	123.650564	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x1cf6 AAAA consumer.cloud.gist.bui
23681	123.650797	10.120.44.171	10.120.44.9	DNS	85	Standard query 0xaaaa A consumer.cloud.gist.build
23682	123.650916	10.120.44.171	10.120.44.9	DNS	85	Standard query 0x0cf4 HTTPS consumer.cloud.gist.bu
23691	123.699973	10.120.44.9	10.120.44.171	DNS	178	Standard query response 0x1cf6 AAAA consumer.cloud
23693	123.711220	10.120.44.9	10.120.44.171	DNS	101	Standard query response 0xaaaa A consumer.cloud.gi
23694	123.711448	10.120.44.9	10.120.44.171	DNS	178	Standard query response 0x0cf4 HTTPS consumer.clou

Domain Name System: Protocol

Packets: 23939 · Displayed: 223 (0.9%)

Profile: Default

4. Task 3: Detailed Packet Analysis

One packet from each of the three filtered protocols was selected for in-depth analysis. The key header details are recorded in the table below.

Protocol	Source IP	Destination IP	TTL	Packet Length	Flags
ICMP	10.120.44.171	8.8.8.8	128	74 bytes	0x00 (Not set)
HTTP	10.120.44.171	93.184.216.34	128	734 bytes	0x18 (PSH, ACK)
DNS	10.120.44.171	8.8.8.8	128	71 bytes	(N/A for UDP)

The screenshot below shows the detailed packet-layer view for the ICMP Echo Request, with the Frame, Ethernet, Internet Protocol, and ICMP layers expanded.

5. Task 4: Custom Filter Creation

A custom filter was created to isolate all packets originating from the host IP (10.120.44.171) and destined for standard web ports (80 for HTTP or 443 for HTTPS).

Filter String:

ip.src == 10.120.44.171 && (tcp.port == 80 || tcp.port == 443)

This filter successfully displayed all outgoing HTTP and HTTPS traffic, which consisted primarily of TCP and TLSv1.2 (Transport Layer Security) packets.

(Insert your screenshot Q4.png here)

6. Summary and Key Insights

This lab provided several key insights into practical network communication.

- **Most Active Protocols:** An analysis of the capture's Statistics > Protocol Hierarchy menu revealed that **TCP** and **UDP** were the dominant transport protocols, carrying **TLS** (for HTTPS), **DNS**, and **HTTP** traffic.
- **Suspicious Traffic:** No suspicious or anomalous traffic was observed during this controlled capture. All traffic was directly attributable to the actions performed (pinging and web browsing).
- **Key Insights:**
 1. **DNS is a Prerequisite:** The capture clearly shows that a DNS query (using UDP) must be sent to resolve a domain name *before* an HTTP/HTTPS connection (using TCP) can be established.
 2. **HTTP vs. HTTPS Encryption:** A major finding was the visible difference between HTTP and HTTPS. Traffic to http://example.com was unencrypted and readable

(e.g., GET /). In contrast, traffic to <https://wikipedia.org> was encrypted as TLS, demonstrating how HTTPS secures web communication by making the data unreadable to an eavesdropper.

3. **ICMP for Diagnostics:** The ping command's use of ICMP Echo Request/Reply packets was clearly observed, providing a practical demonstration of this core network diagnostic tool.

7. Bonus Task: tshark Command-Line Analysis

The tshark command-line utility was successfully used to automatically extract all HTTP request URLs directly from the .pcap file.

Command Used:

```
tshark -r ...\\capture_lab1.pcap -Y http.request -T fields -e http.host -e http.request.uri
```

(Insert your screenshot bonus.png here)

8. Conclusion

This assignment provided valuable hands-on experience in capturing, filtering, and analyzing live network traffic with Wireshark. The tasks successfully demonstrated the layered model of networking and the fundamental protocols (DNS, ICMP, HTTP, TCP/UDP) that power the internet.